Bioactives

PEPHA®-TIGHT CB

Choose the fast lane to superior skin tightening

HALAL FOOD COUNCIL OF EUROPE HALAL

CHINA LISTED

Verified by ECOCERT

COSMOS APPROVED

Raw material verified by ECOCERT GREENLIFE, conform to the COSMOS Standard

100% NATURAL ORIGINAL CONTENT ISO16128 PEPHA®-TIGHT CB

NUTRITION · HEALTH · BIOSCIENCE

DSM

BRIGHT SCIENCE. BRIGHTER LIVING.

The fast and superior

Over time the skin tends to look slack, less toned and less firm. This is the result of a slow, gradual aging process, where the skin's connective tissue becomes increasingly degenerated.

These signs of skin aging may be masked by the use of a skin tightener. However consumers look for more than only tightening. They look for both fast and long-lasting solutions. With PEPHA®-TIGHT CB for the first time it is possible to combine short and long term efficacy with real benefits.

PEPHA®-TIGHT CB is a unique natural tightening and firming active ingredient: it delivers both a **fast**, instant tightening experience, and a **superior** long-term skin firming effect.

PEPHA®-TIGHT CB is 100% naturally-derived according to ISO 16128. It's a highly purified biotechnologically produced extract from the microalgae *Nannochloropsis oculata* combined with a well-balanced fraction of polysaccharides.

Key facts

UNIQUE PRODUCT FEATURES

- Unique combination of purified microalgae active components and a high-performing polysaccharide
- Fast performing with short term and long term efficacy
- Highly controlled fermentation technology in photobioreactors for reliability and optimized composition

BENEFITS

- Instant perceivable tightening experience
- Tightening effect for at least 4 hours after unique application
- Long term skin firming effect after multiple applications

COSMETIC APPLICATION

- Facial skin tightening products
- Protective care for long-term firmer skin
- Firming and lifting Décolleté formulations
- Added value for slimming and anti-wrinkle products

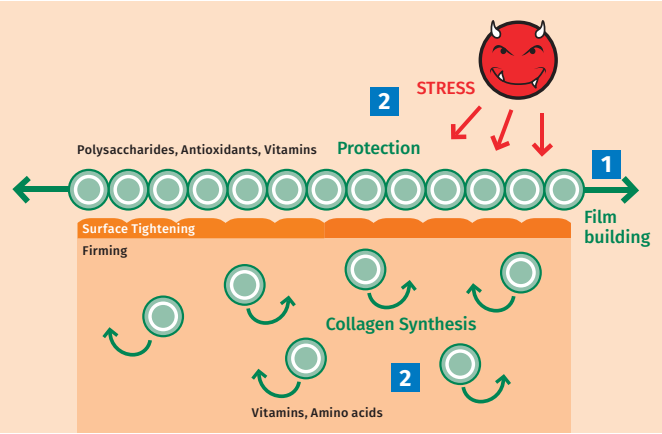
SUGGESTED CONCENTRATION

- 1–5% PEPHA®-TIGHT CB in cosmetic formulations

INCI NAME (ACTIVE)

- *Nannochloropsis oculata* Extract, Pullulan
- Chinese INCI: Algae Extract, Pullulan

Mechanisms

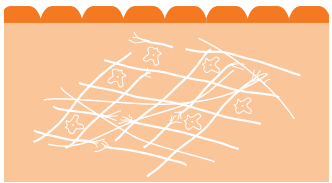


How does PEPHA®-TIGHT CB act on the skin?

1 Short-term effect: PEPHA®-TIGHT CB forms a thin film on the skin, exerting an instant perceptible tightening effect.

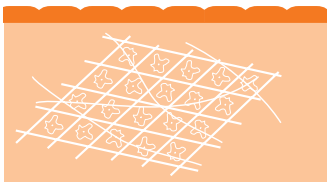
2 Long-term effect: PEPHA®-TIGHT CB acts as a protective shield against oxidative stress, helps to stimulate the formation of collagen I, which is an essential part of the skin's connective tissue, and provides a long-term firming benefit when applied regularly.

If the Extra Cellular Matrix is damaged and disorganized...



...the skin is sagging.

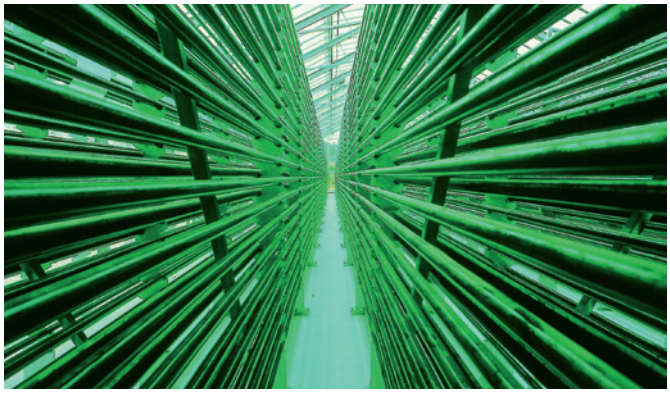
If the Extra Cellular Matrix is intact...



...the skin is tight.

Technology

The microalgae *Nannochloropsis oculata* are cultivated in special photobioreactors, where they are optimally exposed to light and CO2, which are essential nutrients for their proliferation. This unique production technology enables a very pure and standardized product, leading to an high content and consistent quality of active substances, such as amino acids, vitamins, and antioxidants.



Fermentation equipment for the production of the microalgae

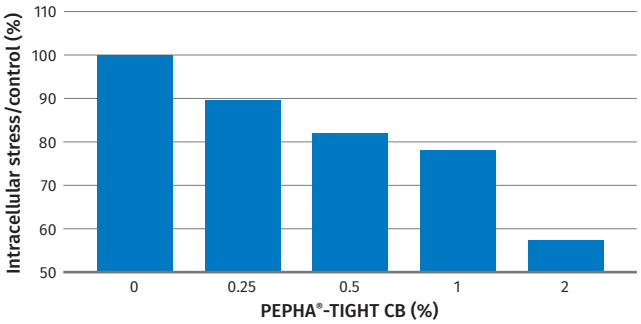
Efficacy

In vitro

Protection against oxidative stress

Reactive Oxygen Species (ROS) are known to accelerate the aging process. In addition, aged cells are less able to protect themselves against oxidative stress. This leads to an accelerated collagen breakdown and cross-linking of proteins. The skin gets thinner and less resilient.

In vitro protection was measured by the ROS assay: Human fibroblasts were exposed to hydrogen peroxide (H2O2) to induce an intracellular oxidative stress. The release of ROS was quantified by fluorescence staining.

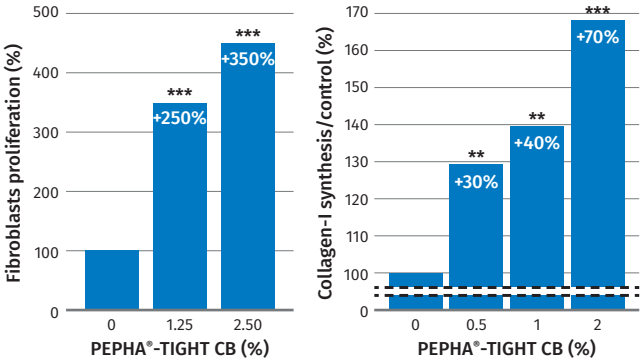


Results: Dose dependent protection of human fibroblasts from oxidative stress, and consequently protection of the collagen fibre network from damaging processes.

Stimulation of fibroblasts proliferation and collagen-I synthesis

Aged skin is characterized by a decline of cell number and proliferative cells. As a consequence, less ECM is produced, leading to a lack of structural components, such as collagen fibres. The skin gets thinner, less resilient and less firm.

High increase of fibroblasts proliferation and of collagen-I synthesis



***p<0.01 against 0. ***p<0.001 against 0.*

Results:

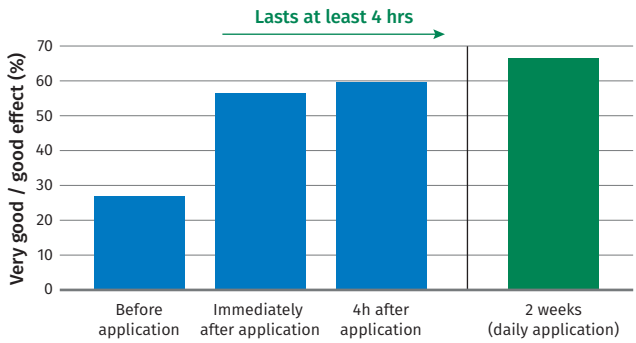
- Increase in cell proliferation of human fibroblasts, and consequently support for collagen production increase.
- Stimulation of collagen-I synthesis in human fibroblasts, improving the structure of the connective tissue.

In vivo

Instant and long-term tightening effects

Clinical studies have shown the instant tightening effect of PEPHA®-TIGHT CB.

In this study, 30 volunteers applied a gel containing 3% PEPHA®-TIGHT CB over a period of 2 weeks.

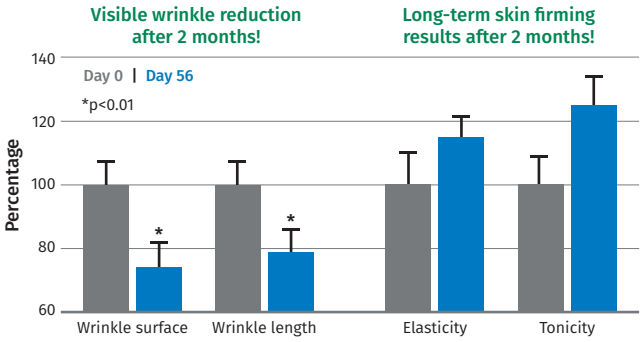


Results:

- Lasts at least 4 hrs after a single application.
- Almost 70% of volunteers quoted as having a good to very good tightening effect after 2 weeks of multiple applications.

Long-term anti-aging effects

12 volunteers applied a cream containing 3% PEPHA®-TIGHT CB over a period of 56 days. Measurement of wrinkles and of skin's elasticity and tonicity were made by silicone replicas and cutometry, respectively.



Results:

- 26% average reduction of wrinkle surface and 21% average reduction of wrinkle length.
- 24% average improvement of skin tonicity and 15% average improvement of skin elasticity.

For more information, please visit www.dsm.com/personal-care

Europe
DSM Nutritional Products Europe Ltd
Personal Care
Wurmisweg 576, CH-4303 Kaiseraugst
Switzerland
Phone: +41 (61) 815 8888
Email: info.pc-emea@dsm.com

Asia Pacific
DSM Singapore Industrial Pte Ltd.
trading as DSM Nutritional Products Asia Pacific
30 Pasir Panjang Road
Mapletree Business City #13-31
Singapore 117440
Phone: +65 6632 6617
Fax: +65 6632 6600
Email: info.pc-apac@dsm.com

North America
DSM Nutritional Products, LLC
45 Waterview Boulevard, Parsippany, NJ 07054
United States of America
Phone: +1 800 526 0189
Fax: +1 973 257 8580
Email: info.pc-na@dsm.com

Latin America
DSM Produtos Nutricionais Brasil S.A.
Av. Juscelino Kubitschek, 1909, Torre Sul, 5o andar
03178-200 Brasil
Phone: + 55 11 3760 6409
Fax: + 55 11 3760 6492
Email: info.pc-latam@dsm.com

China
No. 476, Libing Road, Zhangjiang
Pudong, Shanghai 201203
China
Phone: +86 21 6171 8240
Fax: +86 21 61716266
Email: info.pc-china@dsm.com

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